

A new view on NAmE high negated questions: Experimental evidence and analysis¹

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Abstract. The consensus on English High Negated Questions (HNQs) is that they necessarily convey the speaker's bias for the prejacent. I argue that it is the speaker's bias for the addressee's belief in the prejacent that is necessarily conveyed by HNQs. I support this claim with experimental results, which show that HNQs are not deemed felicitous without the speaker's bias for the addressee's belief in the prejacent even when there is speaker bias for the prejacent, and that they are felicitous when there is speaker bias for the addressee's belief in the prejacent but no speaker bias for the prejacent. I propose a theoretical analysis which combines independently motivated elements to predict this phenomenon and shed further light on two issues that are unresolved in the literature.

Keywords: high negated questions, biased questions, question polarity, decision theory.

1. Introduction

A major function of language is information exchange. Typically, assertions give information, while questions seek it. Biased questions are unique in this respect, as they seek information just like other questions but also convey information about the speaker's knowledge or belief state.

High Negated Questions (HNQs) in English, which are questions with negation syntactically above the subject such as (1), have been observed to mandatorily convey speaker bias.

(1) Isn't John a teacher?

Previous works agree that HNQs obligatorily convey speaker bias, and that this bias is for the question's prejacent, i.e. the propositional content of the sentence radical of the question. For instance, on this view, (1) conveys the speaker's belief that John is a teacher. They differ, however, in their accounts of this phenomenon. Against this backdrop, this study aims for two things. The first is a more accurate characterization of the content of the bias. I challenge the consensus that an HNQ necessarily conveys the speaker's bias for the prejacent and instead propose that it necessarily conveys the speaker's bias for the addressee's belief in the prejacent. Taking (1) again as an example, this means that the bias conveyed is that the speaker expects the addressee to believe that John is a teacher. I provide evidence from an experimental study to support this claim.

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Second, I develop a theoretical account of the bias phenomenon. It draws from extant theories on the semantic denotation of HNQs (Goodhue, 2022), the context-updating conventions of polar questions (Farkas and Bruce, 2010; Farkas and Roelofsen, 2017), and the choice of the polar question form (van Rooij and Šafářová 2003). It utilizes aspects of previous works that have independent motivation other than explaining the bias in HNQs, which, combined, predict that an HNQ conveys the speaker’s bias for the addressee’s belief in the prejacent. I thus provide a more precise description of the phenomenon of mandatory bias arising from HNQs, and a parsimonious theoretical account for this bias.

2. Polar questions in English

This section provides basic information about polar questions in English and discusses the difference between the denotation of HNQs as opposed to other types of polar questions.

2.1. Overview

Polar or “yes/no” questions are questions whose answers are a pair of polar opposites, positive and negative. I refer to polar questions with positive propositional content (e.g., 2a) as Positive Polar Questions (PPQs), those with low negation (e.g., 2b) as Low Negated Questions (LNQs), and those with high negation (e.g., 2c) as High Negated Questions (HNQs).

(2) English polar questions

- a. PPQ: Does Jane live in Paris?
- b. LNQ: Does Jane not live in Paris?
- c. HNQ: Doesn’t Jane live in Paris?

All English polar interrogatives fall into one of these three categories. For this reason, I consider only the PPQ/LNQ/HNQ distinction, although some previous work on biased questions examines finer-grained subcategories within these larger categories. A thorough analysis of the three main types should lead to a better understanding of any subtypes.

2.2. The denotational meaning of PPQs and LNQs

Hamblin (1973) states that the denotation of a question is the set of all possible answers to that question. With polar questions, this is often implemented using the question operator ? (Romero and Han, 2004; Dayal, 2016, a.o.), whose denotation is given in (3a) below.

(3) Denotation of polar questions in Hamblin (1973) semantics

- a. $\llbracket ? \rrbracket = \lambda p_{\langle s,t \rangle} . \lambda q_{\langle s,t \rangle} . [q = p \vee q = \lambda w_s . \neg p(w)]$
- b. $\llbracket ? p \rrbracket = \{p, \neg p\}$
- c. $\llbracket ? \neg p \rrbracket = \{p, \neg p\}$

The question operator ? combines with the proposition p expressed by the sentence radical of an interrogative sentence and yields a partition of worlds of the form $\{p, \neg p\}$. A PPQ with propositional content p has the LF $?p$ and thus has the denotation $\{p, \neg p\}$. Its LNQ counterpart

has the LF $?¬p$ and has the denotation $\{p, ¬p\}$. A PPQ and an LNQ whose prejacent minimally differ in terms of the presence of negation thus have identical semantic denotations, $\{p, ¬p\}$. Following previous authors (van Rooij and Šafářová, 2003; Romero and Han, 2004, a.o.)², I adopt this view regarding PPQs and HNQs.

The denotation of HNQs, on the other hand, poses a more complicated problem. Certain properties of an HNQ suggest that its denotation is different from that of its PPQ or LNQ counterpart. Section 2.3 lays out these distinguishing characteristics of HNQs.

2.3. The denotational meaning of HNQs

This section discusses why the LF of HNQs should be different from that of PPQs/LNQs.

2.3.1. The inner-/outer-negation ambiguity debate

Ladd (1981) claims that HNQs are ambiguous between “inner-negation” and “outer-negation” readings. The former, according to Ladd, requests a reason to believe in the negation of the prejacent. Under this view, *S*’s utterance of an HNQ in (4a) requests a reason to believe that Jane is not coming. Ladd claims, on the other hand, that the latter requests a reason to doubt the prejacent. For example, *S*’s HNQ in (4b) requests a reason to doubt that Jane is coming.

- (4) Inner-/outer-negation readings of HNQs (Romero and Han, 2004: pp.610-611)
- a. Context: Pat and Jane are two phonologists who are supposed to be speaking in our workshop on optimality and acquisition.
A: Pat is not coming. So we don’t have any phonologists in the program.
S: Isn’t Jane coming either? (inner-negation)
 - b. *A*: Ok, now that Stephan has come, we are all here. Let’s go!
S: Isn’t Jane coming too? (outer-negation)

Romero and Han (2004) attribute the inner-/outer-negation ambiguity to whether the prejacent is directly under the scope of negation. According to them, negative polarity items (NPIs) are licensed in the inner-negation reading as shown in (4a), because the prejacent is directly under the scope of negation and hence a downward-entailing environment is formed. In contrast, positive polarity items (PPIs) are licensed in the outer-negation reading, as evidenced in (4b), because the prejacent is not directly negated.

Although it is universally agreed that the outer-negation reading is a valid reading of HNQs, there is debate over the availability of the inner-negation reading. Goodhue (2022) provides compelling evidence that the prejacent is not directly under the scope of negation in HNQs at least in North American English (NAme). Name being the focus of this study, I accept Goodhue’s argument and restrict my attention to the outer-negation interpretation of HNQs.

² See, however, Biezma and Rawlins (2012) for an approach where a polar question denotes a singleton set.

2.3.2. Intervening operator in the LF of HNQs

I focus on the outer-negation reading of HNQs, where the prejacent of an HNQ is not negated. If we aim to adhere to the principle of compositionality, however, we must assume that negation does exist in the LF of an HNQ. To resolve this dilemma, I follow previous approaches (Romero and Han, 2004; Krifka, 2017; Goodhue, 2022, a.o.) in assuming the existence of an intervening operator between the negation and the prejacent.

Such approaches assume that the negation in HNQs scopes over a high operator at the semantic denotation level³. The specific denotation of the operator differs in each account, but the resulting LF and denotation all have roughly the form in (5), where p is the prejacent of the HNQ, and OPER is a stand-in for the specific operator proposed in each account.

(5) The LF and denotation of an HNQ with an intervening operator

- a. LF: $? \neg \text{OPER} p$
- b. Denotation: $\{\text{OPER} p, \neg \text{OPER} p\}$

One consequence is that an HNQ has a different denotation from its PPQ or LNQ counterpart. This property will play a role in deriving the bias encoded in HNQs in Section 4.

3. Empirical characterization of bias in HNQs

I now turn to the bias phenomenon that is unique to HNQs. I first identify the two dimensions of bias that are relevant to the felicity of polar questions, and then present the novel observation that HNQs convey original speaker bias for the addressee's belief in the prejacent. I support this observation with experimental results.

3.1. Two dimensions of bias and the consensus on the bias in HNQs

Two dimensions of bias are identified (Büring and Gunlogson, 2000; Sudo, 2013; Domaneschi et al., 2017, a.o.) as relevant to the felicity of polar questions: original speaker bias (or simply original bias) and contextual evidence bias (or simply contextual evidence). This section briefly characterizes them and restates the consensus on the bias in HNQs in these terms.

A note on the term “bias”: I use this term interchangeably with “belief” or “expectation” throughout this paper, but I do not intend it as meaning doxastic necessity. Rather, if someone thinks a proposition is more likely to be true than false, then I consider them to have bias for this proposition.

Original speaker bias is the belief or expectation that the speaker has before the current conversation. A speaker who has positive original bias regarding a proposition believes that the proposition is more likely to be true than not prior to the ongoing conversation. A speaker who has negative original bias regarding a proposition believes that the proposition is more likely

³ Other works treat this operator as contributing non-denotational, not-at-issue meaning (see Section 3.2 of Romero (2024) for an overview).

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to be false than not. A speaker may also be neutral, i.e. have neither positive nor negative bias, concerning a proposition prior to the conversation.

Contextual evidence bias is the bias generated by evidence that has become available to all discourse participants in the present discourse context. As it is available to all discourse participants, contextual evidence belongs in the Common Ground (CG) (cf. Stalnaker, 1999 p.86). Like original speaker bias, contextual evidence bias can be for, against, or neutral about a certain proposition.

The consensus regarding HNQs is that they require the speaker to have original bias for the prejacent (Romero and Han, 2004; Domaneschi et al., 2017; Goodhue, 2022; Bill and Koev, 2023, a.o.; see also Romero, 2024). This, in turn, means that when an HNQ is uttered, it necessarily signals that the speaker had original bias for the prejacent. Section 3.2 provides observations that challenge this consensus and hypothesizes that the necessary condition for the felicity of HNQs is the speaker's original bias for the addressee's belief in the prejacent.

3.2. Observations and hypothesis

This section presents two observations, each via an example. First, consider a context in which the speaker has original bias for a certain proposition, but does not believe that the addressee believes in that proposition. In such contexts, an HNQ does not seem to be natural as exemplified below.

(6) Type 1

Context: Sean and Gwen are friends. They agreed to meet today inside the school library to do homework together. Gwen said she would be wearing a red shirt and yellow jeans. Sean arrives at the library and searches for Gwen, but he cannot find her. Sean spots the librarian working at his desk. Sean knows that the librarian does not know who Gwen is, and that the librarian probably does not remember every single person who comes into the library every day. Since the librarian is the only other person there, Sean asks:

Sean: Hi, excuse me, #wasn't someone in a red shirt and yellow jeans here today?

Here, Sean has original bias for "Someone in a red shirt and yellow jeans was here today" due to Gwen's previous utterance. However, Sean cannot expect the librarian to believe in this proposition. In this context, the HNQ seems infelicitous.

In the second type of examples, the speaker does not have original bias for a certain proposition but has original bias that the addressee believes in that proposition. In such examples, an HNQ is felicitous as exemplified below:

(7) Type 2

Context: Stanley and Alma are at a racecourse to watch horseracing. Stanley believes that Speedie is the fastest horse there. Stanley knows that Alma thinks differently: she believes that Lightning Bolt is the fastest. At the racecourse, Stanley sees Alma betting on Speedie. Stanley asks:

Stanley: Why are you betting on Speedie? ✓ Isn't Lightning Bolt faster?

Here, Stanley's original bias is that "Lightning Bolt is faster than Speedie" is false. However, Stanley also has original bias that "Alma believes that Lightning Bolt is faster than Speedie". In this case, though the speaker has no original bias for the prejacent, an HNQ is felicitous.

Drawing on these observations, I propose the following hypothesis and predictions, which are tested in the experimental study presented in the next section.

(8) Hypothesis and predictions

- a. **Hypothesis:** The necessary condition for an HNQ to be felicitous is original speaker bias for the addressee's belief in the prejacent, rather than original speaker bias for the prejacent itself.
- b. **Predictions:**
 1. An HNQ will be judged unnatural in Type 1 examples, where the speaker has original bias for the prejacent, but does not have original bias for the addressee's belief in the prejacent.
 2. An HNQ will be judged natural in Type 2 examples, where the speaker has no original bias for the prejacent, but has original bias for the addressee's belief in the prejacent.

3.3. Experiment

3.3.1. Methods

Participants. 100 US-based English-speaking participants were recruited from Prolific Academic. Participants were recruited and provided consent following IRB protocols and were compensated according to Prolific policy (£6/40 min).

Stimuli. All stimuli consisted of a passage providing the context, followed by three possible continuations taking the form of a PPQ, an LNQ, and an HNQ. Participants were asked to rate the naturalness of each continuation on a 1-7 Likert scale. A total of 18 stimuli were used, with 12 target and 6 fillers. 6 of the target stimuli had Type 1 contexts, and the other 6 had Type 2 contexts. The contexts in the target stimuli were intended to represent a range of social relationships (e.g., a student and a librarian, two friends, a person and her friend's son), in order to mitigate any potential effects of politeness or other social factors.

Procedure. Participants were redirected to the experiment's HTML page via a link on the Prolific website. They rated the naturalness of the PPQ/LNQ/HNQ continuation for each context on a Likert scale. The first two stimuli were fixed to be fillers so as to allow participants to practice interacting with the HTML page. The remaining stimuli were presented in random order. The three possible continuations were always presented in the order of PPQ – LNQ – HNQ. Below is an example of an experimental trial.

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Seb is an employee at the customer service team of a company. Today, an intern in his team is dealing with a particularly harshly-worded complaint letter from a client. The intern says to Seb, "I don't understand why this client is reacting this way. There just doesn't seem to be a reason."

Seb, with his many years of experience, knows that when a client is exceptionally upset about a problem, it is usually because the client has encountered a similar problem before. Because the intern is new to the company, Seb does not expect the intern to know this. Also, Seb does not expect the intern to know every detail of the client's history. As a reply to what the intern just said, Seb asks:

Has this client filed a similar complaint before? *

Impossible 1 2 3 4 5 6 7 Completely natural

Has this client not filed a similar complaint before? *

Impossible 1 2 3 4 5 6 7 Completely natural

Hasn't this client filed a similar complaint before? *

Impossible 1 2 3 4 5 6 7 Completely natural

Figure 1: A sample of an experimental trial.

3.3.2. Results

Predictions made in (8b) were generally borne out in the experiment. A visualization of the results is given below.

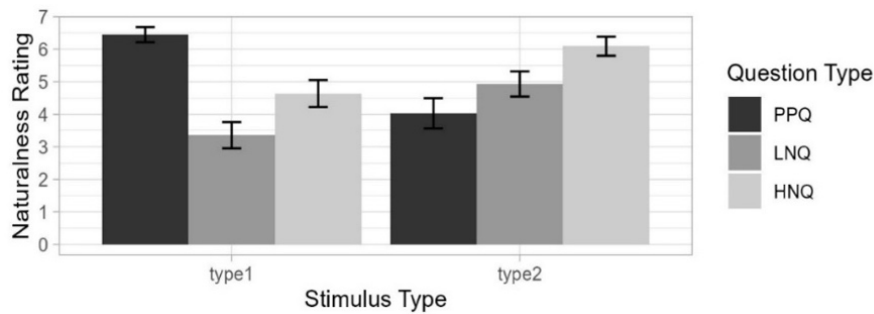


Figure 2: Naturalness rating for PPQ/LNQ/HNQ continuations: mean and 95% confidence interval.

In Type 1 examples, HNQs were judged to be less natural than their PPQ counterparts, which were the most natural continuations in these contexts. In Type 2 examples, HNQs were judged to be the most natural of the three possible continuations⁴. These findings correspond to the first and second predictions made in (8b).

Statistical tests further reveal that these differences are significant. First, paired Wilcoxon signed rank tests report that the mean naturalness rating for HNQs is significantly less than that of PPQs in Type 1 stimuli ($V = 9.5$, $p < 2.2 \times 10^{-16}$), and that the mean naturalness rating for HNQs is significantly greater than that of LNQs ($V = 300$, $p = 1.28 \times 10^{-12}$) and PPQs ($V =$

⁴ The results for each target item show the same tendency as the aggregate results: In all six Type 1 items, the HNQ is rated the second most natural, following the PPQ. Similarly, in all six Type 2 items, the HNQ is rated the most natural, followed by the LNQ in all but one of the items (the by-item plots for both types are available at the URL in footnote 1). I take this as evidence that, if there exists an effect of social factors (e.g., politeness), at least it is not so big as to create a difference in the overall tendency for each item.

333.5, $p = 1.98 \times 10^{-13}$) in Type 2 stimuli. Also, the mean naturalness rating for HNQs is significantly greater in Type 2 stimuli than in Type 1 stimuli ($V = 127.5$, $p = 5.86 \times 10^{-16}$)⁵.

Second, mixed effects linear regression models, with participant and item as random intercepts, report that the continuation question type (PPQ/LNQ/HNQ) has a significant effect on the mean naturalness rating in both Type 1 and Type 2 stimuli. Specifically, in Type 1 stimuli, the continuation being an HNQ had a significant negative effect on the mean naturalness rating ($\beta = -1.810$, $SE = 0.091$, $t = -19.95$, $p < 0.001$). In Type 2 stimuli, the continuation being an HNQ had a significant positive effect on the mean naturalness rating ($\beta = 2.062$, $SE = 0.106$, $t = 19.549$, $p < 0.001$). The statistical tests provide evidence that HNQs are not the most preferred question type in Type 1 contexts, and that they are the most preferred question type in Type 2 contexts, largely in accordance with the predictions in (8b).

4. Proposed analysis

I now propose an analysis of the bias phenomenon of HNQs. As shown in Section 3, the mandatory bias that HNQs convey is the speaker's bias for the addressee's belief in the prejacent. I show that this bias falls out as a necessary consequence of the interaction between the semantics and the discourse properties of HNQs.

4.1. The Interrogativity Principle

As a general principle governing the felicity of the speech act of asking, I put forth the Interrogativity Principle in (9). This is a revision of Goodhue's (2022) Interrogativity Principle in (10). The Interrogativity Principle provides insight into the interaction between the original speaker bias and the contextual evidence bias conditions involved in polar questions.

- (9) Interrogativity Principle
Ask a question Q only if the intersection of your belief set and the context set c does not entail a complete answer to Q .
- (10) Interrogativity Principle (Goodhue, 2022; see also Roberts, 2012 and Büring, 2003)
Ask a question Q only if the context set c does not entail a complete answer to Q .

While Goodhue considers only the context set c in his Interrogativity Principle, I posit that the intersection of the speaker's belief set and the context set c should be considered. The context set c is defined as the intersection of the CG, i.e. the joint commitment of discourse participants (Stalnaker, 1978; Farkas and Bruce, 2010, a.o.). An agent's belief set is the set of possible worlds compatible with the agent's beliefs. I have modified the necessary condition for question-asking to include the belief set to reflect the observation that not only public commitments, but also private beliefs can affect whether a discourse participant asks a question about an issue. I assume that the Interrogativity Principle is at play in all examples that I consider.

⁵ I thank an anonymous SuB30 reviewer for suggesting that I use the Wilcoxon signed rank test, rather than a t-test, for this analysis, since a Likert scale represents ordinal, rather than numerical, values.

The Interrogativity Principle predicts that a question is not licensed when the speaker's original bias and the contextual evidence point in the same direction. A speaker's original bias is part of their belief set. The contextual evidence is in the CG and therefore is part of the context set c . If both biases indicate p , the intersection of the speaker's belief set and c entails p , a complete answer to the question whether p . Thus, asking whether p is not licensed. The Interrogativity Principle thus determines which conditions license the act of asking a question and which do not. This will become useful when I discuss the contextual evidence bias condition of HNQs in Section 5.1.

4.2. Theoretical components of the account for the bias in HNQs

This section presents the theoretical components that I use to derive the bias in HNQs: the LF of HNQs, the context-updating conventions of questions, and an account of the choice of polar question form. I thus delineate the division of labor between semantics and pragmatics in deriving the bias in HNQs, and show how they interact to predict the phenomenon described in Section 3.

4.2.1. Logical form of HNQs

As discussed in Section 2.3.2, we can explain why the prejacent of an HNQ is not directly negated by positing the existence of an operator intervening between the negation operator and the prejacent in the LF of an HNQ. Following Goodhue (2022), I assume that this operator is a doxastic necessity operator where the belief holder is the addressee, represented with the symbol $\Box_A$. The LF of an HNQ with prejacent p is as in (11a) under this analysis. Assuming the same semantic denotation of the question operator $?$ as in (3a), this leads to the question partition in (11b).

(11) The LF and denotation of an HNQ (Goodhue 2022)

- a. LF: $? \neg \Box_A p$
- b. Denotation: $\{\neg \Box_A p, \Box_A p\}$

Asking an HNQ with prejacent p is thus equivalent to asking whether the addressee does not believe that p . Such an HNQ partitions the set of possible worlds into ones in which the addressee believes that p , and ones in which the addressee does not believe that p .

4.2.2. Context-updating conventions

After Farkas and Roelofsen (2017), I assume that when an interlocutor expresses an utterance, its semantic denotation is placed on the Table, the discourse component which keeps track of what is at issue (Farkas and Bruce, 2010). This, according to Farkas and Roelofsen, is a basic convention of use which applies to utterances of all syntactic types. Assuming that the denotation of an HNQ with prejacent p is as in (11b), uttering this HNQ would place the issue $\{\Box_A p, \neg \Box_A p\}$ on the Table.

The key difference between PPQs/LNQs and HNQs is that they place different issues on the Table. As stated in Section 2.2, it is standardly assumed that PPQs have the LF $?p$, and that LNQs have the LF $?¬p$. Both result in the partition $\{p, ¬p\}$, and this is the issue placed on the Table when a PPQ or an LNQ is uttered per Farkas and Roelofsen's basic convention of use. In other words, a PPQ or an LNQ of prejacent p raises the question of whether p or $¬p$ should be added to the CG.

In contrast, an HNQ with the same prejacent has the LF $?¬□_Ap$ and gives rise to the partition $\{¬□_Ap, □_Ap\}$. Following the same basic convention of use, uttering an HNQ places the issue $\{¬□_Ap, □_Ap\}$ on the Table instead. In other words, an HNQ with prejacent p raises the question of whether $□_Ap$ or $¬□_Ap$ should be added to the CG. My account of the data that we saw in Section 3.1 will exploit this distinction between HNQs and PPQs/LNQs.

4.2.3. Choosing the polar question form

All polar questions, whether they are PPQs, LNQs, or HNQs, place a partition containing two cells, one positive and one negative, on the Table, although only one of these cells is explicitly pronounced in the question. This section introduces van Rooij and Šafářová's (2003) decision-theoretic account of which cell is chosen to be pronounced in a polar question, which I will later apply to HNQs to derive their bias for the addressee's belief in the prejacent.

Van Rooij and Šafářová (2003) state that the speaker prefers to pronounce the cell whose utility value is higher than that of its negative counterpart when asking a polar question. In other words, if the utility value of the truth of a proposition q is higher than that of $¬q$, the speaker prefers to ask $?q$ rather than $?¬q$, where q may be a positive or a negative proposition.

Drawing on decision theory, van Rooij and Šafářová derive the utility value of a proposition as in (12). P stands for the agent's probability function, or beliefs about the state of the world, and U stands for the agent's utility function, or preferences regarding the state of the world. (12c) restates the authors' proposal regarding which cell is pronounced in polar questions.

- (12) Utility-based analysis of polar question choice (van Rooij and Šafářová, 2003)
- a. An agent's expected utility of current belief-desire state $\langle P, U \rangle$:

$$EU(P, U) = \sum_w P(w) \times U(w)$$
 - b. Utility value of new information q : $UV(q) = EU(P_q, U) - EU(P, U)$
 where P_q is equal to P conditionalized with q , s.t. $P_q(w) = P(w|q)$
 - c. Pronounced cell in polar questions: If $UV(q) > UV(¬q)$, then prefer to ask $?q$

Van Rooij and Šafářová further claim that if a polar question is used for a purely informative purpose, i.e. when the speaker just wants to know which world is the actual one, then the utility of world w is $\log P(w)$ (cf. van Rooij, 2004). Plugging this into the equations in (12a) and (12b), we obtain the following equivalence:

$$UV(q) > UV(¬q) \leftrightarrow P(q) < P(¬q)$$

In other words, if the speaker believes that q is the less likely of the options q and $¬q$, learning that q is the case has a greater utility value than learning that $¬q$ is the case. This is intuitive:

learning a surprising fact gives one more information than learning an unsurprising one. Plugging the above equivalence into (12c), we arrive at the conclusion that if a speaker believes that q is less likely than its negation, they would prefer to ask $?q$.

A clarification regarding the term “believe” is due. Although van Rooij and Šafářová do not make this explicit, their notion of speaker belief is best understood as referring to what we now know as original speaker bias. In van Rooij and Šafářová’s words, it is the “belief-desire state *before* the new information is grounded” (authors’ own italics) that decides which cell to pronounce. Further, since the focus here is informativity, the flavor of the original speaker bias would be epistemic.

4.3. Derivation of the bias in HNQs

Section 4.2 has laid out the following three theoretical components. First, the LF of an HNQ with prejacent p is $?¬□_Ap$. Second, the utterance of a polar question of the form $?q$ places $\{q, ¬q\}$ on the Table. This means that, in the case of an HNQ, the issue $\{□_Ap, ¬□_Ap\}$ is placed on the Table. Third, the fact that one of the cells in a question partition is pronounced rather than the other indicates that the speaker’s utility value for that cell is greater than that of the other. With these, we can now derive the existence and direction of bias in HNQs.

By uttering an HNQ, the speaker chooses to pronounce $¬□_Ap$ over $□_Ap$. Assuming that the speaker’s purpose in asking the question is to gain information, this signals that the informational utility value of $¬□_Ap$ is greater than that of $□_Ap$. This, as I showed in Section 4.2.3, is equivalent to the speaker’s original epistemic bias being such that $P(¬□_Ap) < P(□_Ap)$. In other words, the utterance of an HNQ with prejacent p signals that the speaker had original bias that the addressee believed in p . This conclusion is identical to the empirical generalization that I made in Section 3.

This account has the advantage of using independently motivated theoretical components, none of which is needed solely to account for the bias in HNQs. The intervening operator in the LF of HNQs is needed to account for the syntactic observation that the prejacent is not negated in HNQs. The discourse structure model proposed by Farkas and Bruce (2010) and the conventions of use surrounding it proposed by Farkas and Roelofsen (2017) figure in accounts of how discourse operates in general and are not specific to HNQs. Lastly, van Rooij and Šafářová’s (2003) proposal regarding the pronounced cell of polar questions applies to polar questions in general. Previous accounts such as Romero and Han (2004) and Frana and Rawlins (2019) have faced the criticism that they posit a Principle of Economy whose sole purpose is to dictate the usage of the VERUM operator. The present account can accurately predict the phenomenon without needing such stipulations.

4.4. Pragmatic inference of original speaker bias for the prejacent in HNQs

I have argued that the mandatory bias in HNQs is the speaker’s original bias for the addressee’s belief in the prejacent. A question, then, is why the literature so consistently takes the mandatory bias in HNQs to be the speaker’s original bias for the prejacent. I propose that the speaker’s original bias for the prejacent can arise as a pragmatic implication from many

contexts that allow an HNQ. Crucially, this implication is cancelable since it stems from interlocutors' general expectations about the world, unlike original speaker bias for the addressee's belief in the prejacent. In this section, I first point out an assumption underlying most conversations, then show how this results in the speaker's bias for the prejacent coinciding with the speaker's bias for the addressee's belief in the prejacent in many real-world scenarios.

4.4.1. A remark on conversation in general

A major purpose of a conversation is to increase our knowledge of the world. More technically, we aim to expand the CG, or, equivalently, to decrease the context set. Ideally, by talking to another person, we will be able to pinpoint the actual world out of the myriad possible worlds. For this to work, an interlocutor must operate with a crucial underlying assumption: that she and other conversation participants all live in the same world, which is the "actual" one. This is why, for instance, a dependent commitment (Gunlogson, 2008) is possible: a speaker may use another agent's testimony that p as a source of their own commitment to p .

Due to this basic assumption that one's actual world is the same as other people's, one does not by default expect others to have beliefs that contradict their own beliefs. This is formulated below:

- (13) Let A and B be interlocutors.
- a. If $\Box_A p$, then $\Box_A \neg \Box_B \neg p$ by default. (non-contradiction)
 - b. If $\Box_A \neg \Box_B \neg p$ and A is in a position to assume B 's opinionatedness on p , then $\Box_A \Box_B p$. (opinionatedness)

(13a) states that if an agent A believes that p , they believe that B does not believe that $\neg p$, i.e. that B does not harbor a belief that contradicts theirs. (13b) states that if A is further in a position to assume that B is opinionated whether p , then this belief strengthens to the belief that B believes that p . This can be thought of as the rationale behind Walker's (1996) Collaborative Principle, which states that interlocutors must provide evidence of a detected discrepancy in belief as soon as possible. There is no similar requirement to provide evidence of a detected alignment in beliefs. This indicates the special, non-canonical status of belief discrepancy.

(13) is a general observation about the world that influences how people use language. As such, it also allows exceptions. The account below uses this observation to explain how an HNQ with prejacent p may convey speaker bias for p , as well as when this implication is canceled.

4.4.2. When original speaker bias for p and original speaker bias for $\Box_A p$ coincide

Suppose that a speaker S believes that p , and encounters contextual evidence for $\neg p$. Suppose further that S has no reason to expect the addressee A 's belief to differ from their own, and that S expects A to be opinionated regarding the issue $\{p, \neg p\}$. Thus, the condition in (13b) is satisfied, and S believes that A also believes that p . In other words, to S , $P(\Box_A p) > P(\neg \Box_A p)$. This is a situation where S would have the motivation to ask an HNQ.

A new view on NAmE high negated questions

Most examples with HNQs in the literature depict contexts where the two assumptions in (13) can be safely thought to hold, as in example (14), taken from Goodhue (2022):

- (14) *S* expects her roommate Moira to be home, but she looks everywhere and can't find her. *S* finds **their mutual roommate** *A* in the last room she checks. *S* says to *A*:
S: Isn't Moira home? (Goodhue, 2022: p.381 (3); boldface my own)

Based on real-world knowledge, one can usually assume that one's roommate shares one's knowledge regarding the whereabouts of a third roommate. So, *S* simultaneously expects her roommate Moira to be home and *A* to expect Moira to be home. Crucially, it is the latter bias that licenses the use of an HNQ in this scenario, although in this case the latter bias happens to have been induced from the former. The correlation between the two biases is not a logical necessity, and the next section discusses cases where they diverge.

4.4.3. When original speaker bias for *p* and original speaker bias for $\Box_A p$ diverge

For a speaker's bias for *p* to lead to their bias for the addressee's belief in *p*, the two assumptions in (13) need to hold. If either assumption fails, speaker bias for *p* does not translate to speaker bias for the addressee's belief in *p*.

Example (7), our Type 2 example, is a case where the non-contradiction assumption in (13a) does not hold. The context is such that Stanley, the speaker, knows prior to the conversation that his opinion on whether Lightning Bolt is the fastest and Alma's opinion on that matter are different. This is a case where the speaker has bias for the addressee's belief in the prejacent but not because he himself believes that the prejacent is true. Here, the HNQ is felicitous despite the prejacent being in the opposite direction of Stanley's own belief, unlike in Goodhue's example in (14).

Example (6), our Type 1 example, is where the opinionatedness assumption in (13b) does not hold. In this context, the speaker Sean cannot reasonably expect the librarian to know whether a particular person was in the library. Sean has original bias for the prejacent 'There was someone in a red shirt and yellow jeans here today' and may further have the non-contradiction assumption. World knowledge, however, blocks the opinionatedness assumption, and so Sean's bias for the prejacent does not result in his bias for the addressee's belief in the prejacent. This in turn results in the infelicity of an HNQ in this example.

5. Further puzzles solved

This account also provides insight into some points of contention in the literature, namely, the strength of the speaker's bias toward the prejacent of an HNQ, and the contextual evidence condition of an HNQ.

5.1. Strength of the speaker's bias toward the prejacent

Previous works, which agree that the mandatory bias in HNQs is the speaker's bias for the prejacent, differ as to whether this bias amounts to doxastic necessity or is weaker. These conflicting observations are problematic for accounts that view speaker bias for the prejacent as the core defining characteristic of HNQs. The present account, in contrast, does not face the same problem, and in fact predicts such variability in strength of the speaker's bias for the prejacent.

Goodhue (2022) characterizes the speaker's bias for the prejacent in HNQs as the speaker's doxastic necessity, comparable to the degree of belief signaled by an assertion of the prejacent. He motivates this proposal using the following example:

- (15) Context: *A* and *B* checked the weather forecast when they woke up, and it said it would be a nice day. A couple hours later, they look out the window and see bad weather. *B* says, "It's gross out." *A* replies,
- a. *A*: I know. The forecast said it would be nice out.
 - b. *A*: I know. Didn't the forecast say it would be nice out? (Goodhue, 2022: p.403 fn. 28)

Goodhue, observing that the assertion in (15a) and the HNQ in (15b) are interchangeable, argues that "the same sorts of evidence that license assertion also support the use of HNQ. (p.402 fn.28)"

Bill and Koev's (2023) account, in contrast, predicts that the speaker bias for the prejacent in HNQs is weak. As support, they provide experimental results where 78% of the participants chose a "weak follow-up" to an HNQ. A sample question from their experiment is given below:

- (16) Doesn't Paul own a bike? Because...
- a. I thought he did. (weak follow-up)
 - b. I was sure he did. (strong follow-up)

Bill and Koev (2023) argue that the preference for (16a), which encodes weak belief, over (16b), which encodes certainty, indicates that HNQs convey a weak bias for the prejacent. This conflicts with Goodhue's (2022) characterization.

Such conflicting observations on speaker bias strength for the prejacent of HNQs are unproblematic, and, in fact, expected under my analysis. I argued in Section 4.4 that speaker bias for the prejacent in HNQs arises from a pragmatic inference dependent on contextual factors. As such, it is natural that its strength should vary depending on the context.

To see how an inference of a weak speaker bias for the prejacent may arise, consider (17), a context in which Bill and Koev's example "Doesn't Paul own a bike?" can occur felicitously with the weak follow-up "Because I thought he did.":

- (17) Context: *A* has previously told *S* that Paul owns a bike. *S* doesn't know Paul all that well and took *A*'s word for this matter. Today, *A* says, "John and Tam are the only people I know who own a bike." To this, *S* replies:
S: Doesn't Paul own a bike? Because I thought he did.

Here, *S*'s belief in 'Paul owns a bike' is dependent on *A*'s prior utterance. As Faller (2019) notes, such a dependent commitment is weaker than that of a standard assertion. Thus, the weak follow-up "I thought he did." is felicitous, while the strong "I was sure he did." is less so.

In contrast, in Goodhue's (2022) example in (15), *A* and *B* are both sources, in Gunlogson's (2008) sense, of the information that the weather forecast said it would be nice out. Furthermore, *A* and *B* both know that they are joint sources to this information. So, in this case, *A* is both certain that *p*, and that *B* is certain that *p*, where *p* is the proposition "the forecast said it would be nice out". *A*'s own belief in *p* licenses *A*'s assertion in (15a), and *A*'s belief in *B*'s belief in *p* licenses *A*'s asking an HNQ in (15b). Goodhue's examples represent a special case where the speaker just happens to have both the belief in the prejacent and the belief that the addressee believes in the prejacent, and we cannot draw the generalization that HNQs and assertions require the same types of evidence solely from such examples.

Finally, as discussed in Section 4.4.3, the inference that the speaker believes in the prejacent of an HNQ can even be canceled: example (7) shows that a speaker who believes in the negation of the prejacent can felicitously utter an HNQ. The crucial factor in Goodhue's example in (15), Bill and Koev's example in (16), and my example (7) alike is the speaker's bias that the addressee believes in the prejacent, and the speaker's own bias for the prejacent may vary in strength.

5.2. Contextual evidence bias condition

There is also a lack of consensus concerning the contextual evidence bias condition on HNQs. Table 1 summarizes the varying predictions on this matter. Note that all accounts characterize this condition in terms of whether *p* (= prejacent of the HNQ).

	Büring and Gunlogson (2000)	Trinh (2014); Northrup (2014)	Goodhue (2023)
For <i>p</i>	#	#	✓
Neutral	✓	#	✓
Against <i>p</i>	✓	✓	✓

Table 1: Proposed contextual evidence conditions for HNQs (adapted from Romero, 2024: p.296 Table 4)

My account sheds light on why such varying predictions arise. As discussed in Section 4.1, the Interrogativity Principle dictates that the contextual evidence needed for a question to be felicitous depends on the original bias that the speaker has. I have argued that an HNQ obligatorily conveys original speaker bias for $\Box_A p$. It therefore follows that its contextual evidence condition should be characterized in terms of $\Box_A p$ rather than *p* as in prior work. Further, since the Interrogativity Principle dictates that original speaker bias and contextual evidence bias cannot be in the same direction, the contextual evidence for an HNQ must be neutral or negative with respect to $\Box_A p$.

Since an HNQ can felicitously occur with any original speaker bias regarding p , I predict that it is also compatible with any contextual evidence regarding p , as long as there is no positive contextual evidence for $\neg_A p$. It may, however, be that in most realistic scenarios, the contextual evidence points in the same direction for p and $\neg_A p$. That is, there is a loose correlation between evidence for p and $\neg_A p$. It is logically possible to felicitously utter an HNQ when there is contextual evidence for p , as Goodhue (2023) proposes. Such a context, however, must also provide evidence neutral towards or against $\neg_A p$, and such a combination of evidence can turn out to be odd in terms of our world knowledge. I suggest this as an explanation for Büring and Gunlogson (2000), Trinh (2014), and Northrup's (2014) proposal that HNQs cannot occur with positive contextual evidence for p .

6. Conclusion

This study provides two main contributions to our understanding of HNQs. The first is improving the characterization of the bias that arises from them. The consensus is that HNQs obligatorily convey speaker bias for the prejacent. I claim that it is in fact the speaker's bias that the addressee believes in the prejacent that is necessary for an HNQ to be felicitous. The results from an experimental study support this novel empirical characterization.

The second contribution of this study is developing a theoretical account for this phenomenon. I delineate the division of labor among the semantic denotation of HNQs (Goodhue, 2022), the context-updating conventions of polar questions (Farkas and Bruce, 2010; Farkas and Roelofsen, 2017), and the speaker's utility-driven choice of question form (van Rooij and Šafářová, 2003) to derive the speaker's bias for the addressee's belief in the prejacent. Combining such independently motivated elements, I parsimoniously account for the bias phenomenon of HNQs and further predict that the speaker's bias for the prejacent can optionally arise as a cancelable implication from an HNQ.

This account has the additional advantage of providing insight into two further issues. The first is the issue of the strength of the speaker's bias for the prejacent. Previous accounts have disagreed on whether this bias amounts to doxastic necessity. Such varying predictions are reconciled under my account, which claims that the speaker's bias for the prejacent is an ancillary implication variable in strength and even cancelable. The second is the contextual evidence bias condition of HNQs. I claim that the present dissonance is due to this condition being characterized in terms of the prejacent and propose instead to characterize it in terms of the addressee's belief of the prejacent. The relationship between the speaker's original bias and the contextual evidence bias, mediated by the Interrogativity Principle, makes such a characterization natural.

HNQs are a special breed of questions: they seek information from the addressee but also give information about the speaker's beliefs, complicating the distinction of information-seeking and information-giving acts. By looking closely into what bias an HNQ conveys and how such a bias arises, this study will hopefully lead to a better understanding of biased questions and information exchange via language in general.

References

- Biezma, M. and K. Rawlins (2012). Responding to alternative and polar questions. *Linguistics and Philosophy* 35(5), 361–406.
- Bill, C. and T. Koev (2023). Question bias from polarity focus. Unpublished manuscript, Leibniz-Center General Linguistics (ZAS) and Univ. of Konstanz.
- Büring, D. (2003). On D-trees, beans, and B-accents. *Linguistics and Philosophy* 26(5), 511–545.
- Büring, D. and C. A. Gunlogson (2000). Aren't positive and negative polar questions the same? Unpublished manuscript, University of California Los Angeles/University of California Santa Cruz.
- Dayal, V. (2016). *Questions*. Oxford: Oxford University Press.
- Domaneschi, F., M. Romero, and B. Braun (2017). Bias in polar questions: Evidence from English and German production experiments. *Glossa: a journal of general linguistics* 2(1):26. 1-28.
- Faller, M. (2019). The Discourse Commitments of Illocutionary Reportatives. *Semantics and Pragmatics* 12(8), 1–53.
- Farkas, D. and K. Bruce (2010). On reacting to assertions and polar questions. *Journal of Semantics* 27(1), 81–118.
- Farkas, D. and F. Roelofsen (2017). Division of labor in the interpretation of declaratives and interrogatives. *Journal of Semantics* 34(2), 237–289.
- Frana, I. and Rawlins, K (2019). Attitudes in discourse: Italian polar questions and the particle mica. *Semantics and Pragmatics* 12(16), 1–55.
- Goodhue, D. (2022). Isn't there more than one way to bias a polar question? *Natural Language Semantics* 30(4), 379–413.
- Goodhue, D. (2023). The evidential condition on asking polar questions. Invited talk at the International Workshop on the Semantics of Non-Canonical Questions, Toronto, Canada, May 17–18.
- Gunlogson, C. (2008). A question of commitment. *Belgian Journal of Linguistics* 22, 101–136.
- Krifka, M. (2017). Negated polarity questions as denegations of assertions. In C. Lee, F. Kiefer, and M. Krifka (Eds.), *Contrastiveness in information structure, alternatives and scalar implicatures*, pp. 359–398. Berlin: Springer.
- Ladd, D. R. (1981). A first look at the semantics and pragmatics of negative questions and tag questions. In R. A. Hendrick, C. S. Masek, and M. F. Miller (Eds.), *Papers from the seventeenth regional meeting of the Chicago Linguistic Society*, pp. 164–171.
- Northrup, O. (2014). Grounds for commitment. Ph. D. thesis, University of California, Santa Cruz.
- Roberts, C. (2012). Information structure in discourse: Towards an integrated formal theory of pragmatics. *Semantics and Pragmatics* 5(6), 1-69.
- Romero, M. and C.-H. Han (2004). On Negative Yes/No Questions. *Linguistics and Philosophy* 27(5), 609–658.
- Romero, M. (2024). Biased polar questions. *Annual Review of Linguistics* 10(1), 279-302.
- van Rooij, R. (2004). Utility, informativity and protocols. *Journal of Philosophical Logic* 33, 389–419.
- van Rooij, R. and M. Šafařová (2003). On polar questions. In R. B. Young and Y. Zhou (Eds.), *The proceedings of Semantics and Linguistic Theory* Vol. 13, pp. 292–309.

- Sudo, Y. (2013). Biased polar questions in English and Japanese. In D. Gutzmann and H.-M. Gärtner (Eds.), *Beyond expressives: Explorations in use-conditional meaning*, pp. 275–295. Leiden/Boston: Brill.
- Stalnaker, R. (1978). Assertion. In P. Cole (Ed.), *Syntax and Semantics 9: Pragmatics*, pp. 315–332. New York: Academic Press.
- Stalnaker, R. (1999). Assertion. *Context and Content*, pp. 78–95. Oxford: Oxford University Press.
- Trinh, T. (2014). How to ask the obvious: A presuppositional account of evidential bias in English yes/no questions. In C. Luka and U. Sauerland (Eds.), *The art and craft of semantics: A festschrift for Irene Heim* Vol. 2, pp. 227–249. Cambridge: MITWPL.
- Walker, M. A. (1996). Inferring acceptance and rejection in dialog by default rules of inference. *Language and Speech* 39(2–3), 265–304.